AI Roadside Tree & Power-Line Risk Mapping for Sri Lankan Roads

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Introduction

Sri Lanka needs a cheap way to find dangerous roadside trees before they fall or grow into power lines. In October 2023 the State Minister of Defence ordered officials to remove trees at risk of falling on both sides of roads (Daily Mirror). Readers responding to that report asked for regular monitoring and expert assessment instead of hasty felling.

This project builds an AI system that turns ordinary smartphone video, filmed from a moving vehicle, into a map of roadside trees. For every tree it estimates two risks: falling, and branches coming too close to overhead wires. Councils and electricity utilities can then trim or inspect only the risky trees and keep the healthy ones.

The system is designed for Sri Lankan roads: narrow, winding and bumpy, with power, telephone and fibre cables often sharing the same poles. These conditions make the problem harder than in earlier studies, and they are the main source of the project's novelty.

Problem Statement

There is no low-cost way to survey every roadside tree in Sri Lanka and the wires around it, so risky trees are found late or felled in bulk.

- **Storms expose the gap.** In one 24-hour period in May 2025, 50,000 electricity breakdowns were reported nationwide during monsoon rain and strong winds, while fallen trees blocked major roads in Colombo (Daily FT).
- **Existing AI answers the wrong question.** The closest camera-based study, from Brazil, only classified street photos as tangled with wires or not, at 74.6% accuracy (Oliveira et al., 2022). A trimming crew needs to know how close each branch is, not just yes or no.
- **Most power-line research flies.** Inspection studies mainly use drone or aircraft images, or LiDAR on vehicles (Liu et al., 2020; TTPLA), which cost more to collect than video from a phone in a car.
- **Models trained abroad may not transfer.** Sri Lankan roads are narrower and bumpier, and power cables share poles with telephone and fibre lines. A model trained elsewhere can confuse cable types or fail on shaky video; this project measures that

gap.

Research Questions and Objectives

The project tests whether phone video from a moving vehicle can replace a tree-by-tree survey on Sri Lankan roads.

Research questions

1. How accurately can AI detect roadside trees and overhead cables in smartphone video from Sri Lankan roads, and tell power cables from telephone and fibre lines?
2. How accurately can the gap between a branch and the nearest wire be estimated from ordinary video, compared with measurements taken on site?
3. Can signs visible from the road (lean, dead branches, trunk defects, species) predict an expert's tree-risk rating?
4. How much accuracy do models trained on foreign data lose on Sri Lankan roads, and how much local data closes that gap?

Main objective: build and test a system that maps every roadside tree on a route with two risk scores: falling, and branch-to-wire clearance.

Specific objectives

- Collect and label the first Sri Lankan street-level dataset of roadside trees and overhead cables, with GPS positions and cable types.
- Train models for tree detection, cable segmentation and cable-type classification.
- Estimate branch-to-wire clearance in metres from video and validate it against on-site measurements.
- Build a tree-risk model and check it against expert assessments.
- Deliver a map dashboard that lists trees to trim or inspect, in priority order.

Literature Review and Research Gap

Earlier work classifies tangled trees, uses fixed stereo cameras on towers, compares tree height with circuit height, or measures the gap with LiDAR on vehicles. We found none that measures branch-to-wire clearance from moving phone video, and none from Sri Lanka.

Study	Images	What it does	Gap for this project
Silva et al., 2026	Satellite tiles (about 0.6 m/pixel) plus Mapillary and KartaView street photos	Maps vegetation touching distribution lines using utility GIS data; a YOLOv8 model trained on 100 street images estimates tree height against circuit height	No clearance distance; quantitative validation left for future work
Oliveira et al., 2022 (Brazil)	1,001 street-level photos, São Paulo and Porto Alegre	Classifies trees as tangled with power or communication lines; 74.6% accuracy	Yes/no only; no distance, no video, no map, no fall risk
Rong et al., 2020	Stereo cameras fixed on transmission towers, about 13 m high	Finds trees and lines in 3D and checks the height gap against clearance rules; tree height error 0.40 m	Needs fixed cameras on each tower; tested at two sites
Bahreini et al., 2024	Mobile LiDAR point clouds (Toronto-3D, about 1 km of street)	Finds the nearest power-line point for each tree point and flags trees within a 1 m safety threshold	Needs LiDAR; distance accuracy not reported; no fall risk
Joshi et al., 2025	Consumer dashcam video from a moving vehicle	Maps roadside trees and poles with GPS triangulation and a corrected monocular depth model; tree height error 2.09 m	No wires or clearance; the closest low-cost pipeline
Arevalo-Ramirez et al., 2024	Street-view images	Classifies tree genera and segments trees and trunks to help arborists screen trees	Finds automatic tree evaluation little studied; no wires, no risk score
Chiu et al., 2023 (WireSegHR)	Ordinary photographs	First wire-segmentation benchmark, built for removing wires in photo editing	Not about trees or distances; useful for pre-training

Study	Images	What it does	Gap for this project
Abdelfattah et al., 2020 (TTPLA)	1,100 aerial images, 3840×2160	8,987 labelled transmission towers and power lines	Aerial view; not street level
Liu et al., 2020	UAV images	Reviews deep learning for power-line inspection	Aerial view; does not cover vegetation
Guruge and Herath, 2026 (USJ)	Ground survey, GIS and LiDAR, Colombo	Maps and scores risky roadside trees in two Colombo divisions	No wires and no video; shows local demand for tree-risk maps
Sri Lankan repositories (UoM, SLIIT, UCSC, Peradeniya, KDU, Ruhuna, SJP), searched October 2026	—	No AI work found on roadside tree risk or tree–wire conflicts	Open gap

This project's contribution: clearance in metres instead of yes/no, from cheap phone video without fixed cameras, drones or vehicle LiDAR, placed on a GPS map, combined with fall risk, and tested on Sri Lankan roads with mixed cables.

Sri Lankan Road Challenges

Sri Lankan roads make every step harder, and handling them is part of the research contribution. They are also why a model trained on Brazilian or European streets is unlikely to work unchanged here (research question 4).

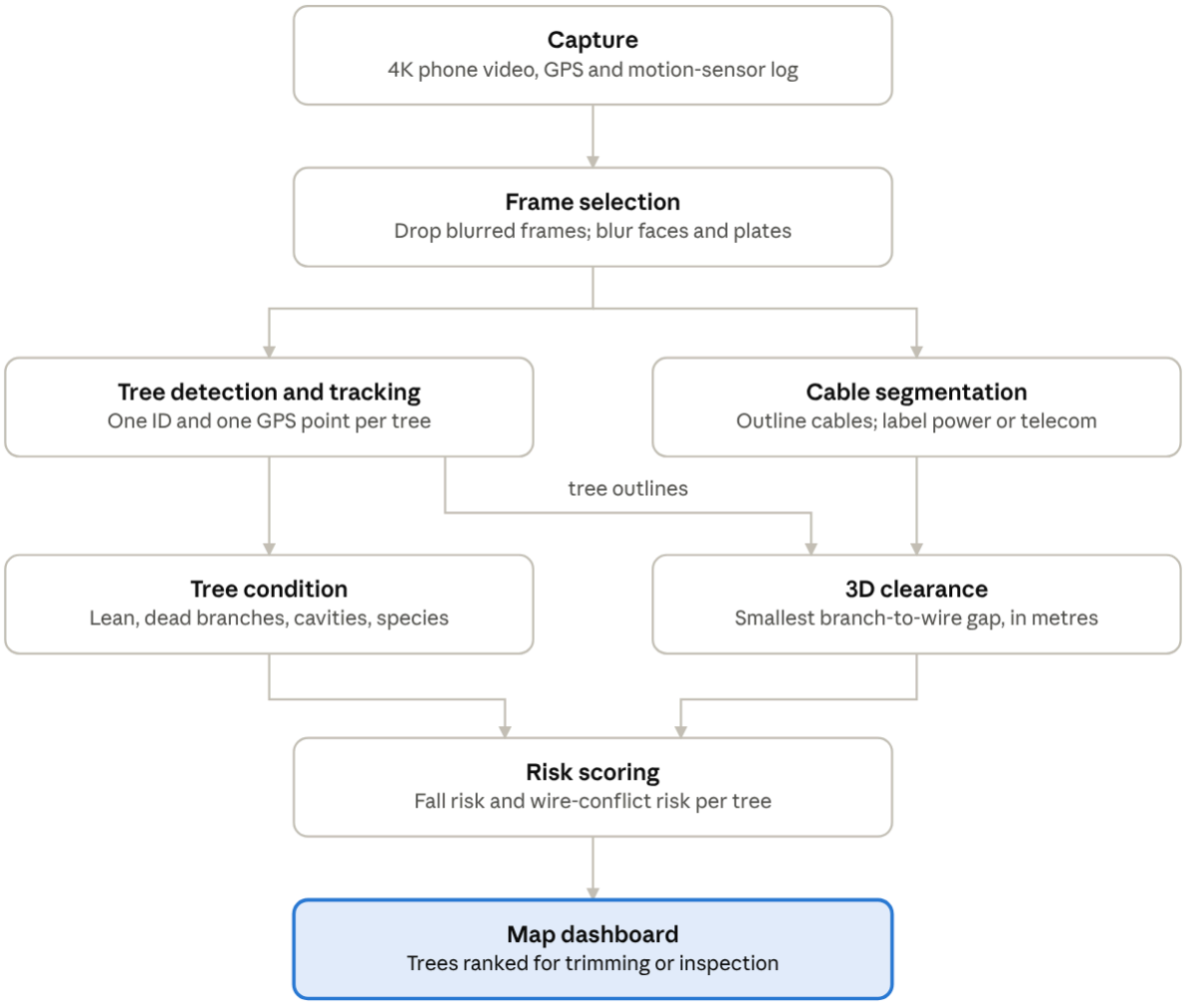
Challenge	Why it matters for the AI	How the project handles it
Bumpy roads and potholes	Shaky, blurred frames hide thin wires	Phone stabilisation on, damped mount, 20–30 km/h, keep only sharp frames, log the phone's motion sensors
Narrow, winding roads with trees at the edge	Trees and wires sit very close and overflow the frame	Ultra-wide lens tilted upward; film each road in both directions

Challenge	Why it matters for the AI	How the project handles it
Power, telephone and fibre cables on the same poles	The model can flag the wrong cable	A cable-type class (power, telecom, unknown); pole fittings such as insulators as clues; LECO or CEB staff confirm labels on sample routes
Buses, three-wheelers and parked vehicles	They hide trees for several frames	Track each tree across frames; repeat routes at quieter times
Deep shade next to bright sky	Wires against the sky and trunks in shade need different exposure	Film mid-morning or late afternoon, use HDR video, avoid filming into the sun
Monsoon rain	Drops on the lens and low light	Film in dry spells; keep a small wet-weather test set to report robustness
GPS drift under canopy and in hills	Trees land on the wrong spot on the map	Snap positions to the OpenStreetMap road line; merge repeat sightings of the same tree

Methodology

One drive becomes a ranked list of trees: trees and cables are found separately, then combined to measure clearance and score risk.

Trees and cables are found separately, then combined into two risk scores per tree



AI pipeline · 8 stages, 2 parallel lanes

Clearance needs both lanes: tree outlines from the left and cable outlines from the right.

Data collection

- **Routes:** target about 100 km, mixing urban roads near the University of Moratuwa with suburban and hill-country roads.
- **Capture:** 4K video at 30 fps from a phone on a damped mount, with GPS and motion-sensor logs; each road filmed in both directions.
- **Ground truth:** for a sample of trees, measure branch-to-wire gaps from the roadside with a total station or a laser rangefinder in two-point (missing-line) mode, because the gap is a 3D distance between two points. An expert rates the same trees' fall risk using a standard visual checklist.
- **Privacy:** blur faces and number plates before any video is stored.

Labelling

- Outline trees, trunks and cables on selected frames; label cable type (power, telecom, unknown) and visible defects (dead branches, cavities, lean, trunk damage visible from the road).
- Use a segmentation assistant such as SAM 2 to speed up outlining; a second member checks a share of every batch.

AI pipeline

1. **Frame selection:** drop blurred frames using a sharpness score and the motion-sensor log.
2. **Tree detection and tracking:** a detector such as YOLO finds trees; a tracker links them across frames, so each tree gets one ID and one GPS position.
3. **Cable segmentation:** a thin-structure segmentation model, pre-trained on WireSegHR and fine-tuned on Sri Lankan frames, outlines cables. WireSegHR only separates wire from background, so a new classifier, trained on local labels and using pole and insulator context, labels each cable power or telecom, bare or insulated.
4. **Clearance estimation:** the moving phone sees each tree from many angles. Structure-from-motion (COLMAP) recovers the camera path and the tree's 3D shape, with a monocular depth model filling in the crown. Wires are too thin for depth models, so each segmented wire is triangulated across frames and fitted with a sagging curve between poles. Scale comes from the GPS track and objects of known size, such as poles and lane widths. Output: the smallest branch-to-wire gap, in metres.
5. **Tree condition:** models estimate lean angle, find dead branches and trunk defects visible from the road, and identify common roadside species. Hidden problems such as root rot are out of scope: the score screens visible defects, as an expert's visual check does.
6. **Risk scoring:** condition, species, clearance and location combine into two scores, fall risk and wire-conflict risk, checked against expert ratings.
7. **Dashboard:** a web map shows every tree with its scores and photos, plus a priority list for trimming or inspection.

Evaluation Plan

Every part is tested on routes held out from training, one per road type, so the results show performance on roads the system has never seen.

What is tested	Metric	Compared against
Tree detection and tracking	Precision, recall, mAP; tracking ID switches per km	Model trained on public data only vs fine-tuned on Sri Lankan data

What is tested	Metric	Compared against
Cable segmentation	IoU and F1 on cable pixels	WireSegHR-only model vs fine-tuned model (research question 4)
Cable type	Accuracy per class (power, telecom)	Labels confirmed by LECO or CEB staff on sample routes
Branch-to-wire clearance	Mean absolute error in metres vs laser measurements; share of trees correctly flagged below a clearance threshold agreed with the utility	2D overlap baseline: does a branch touch a wire in the image?
Fall-risk score	Agreement with expert ratings (weighted kappa); ranking of the riskiest trees	Simple rule-based checklist score
Tree position	Error in metres between mapped and true position	Raw phone GPS
Whole system	Survey time per km; usefulness rating from council or utility staff	Manual survey of the same route

Expected Outcomes and Contributions

The project delivers a dataset, a method and a working tool, each new for Sri Lanka.

1. **Dataset:** the first Sri Lankan street-level dataset of roadside trees and overhead cables, with GPS positions, cable types, defect labels and measured clearances.
2. **Method:** a way to estimate branch-to-wire clearance in metres from moving phone video, tested against on-site measurements.
3. **Evidence:** a measured answer to how well foreign-trained models work on Sri Lankan roads, and how much local data closes the gap.
4. **Tool:** a web map that ranks trees for trimming or inspection, built for municipal councils, the Road Development Authority, CEB and LECO.
5. **Publication:** a conference paper on the dataset and the clearance method.

Tools, Equipment and Data Sources

No DSLR is needed: apart from a phone mount and a laser rangefinder, everything is equipment the team already owns or free software.

Type	Item	Purpose
Equipment	Team smartphones (4K video, GPS)	Main camera and GPS logger
Equipment	Damped phone mount for a car or three-wheeler	Steadier video on bumpy roads
Equipment	Second phone mounted alongside (optional)	Stereo pairs for better depth
Equipment	Laser rangefinder	3D branch-to-wire gaps from the ground, using two-point mode; a total station from the surveying lab is the alternative
Compute	Google Colab GPU and team laptops	Training and testing
Software	Python, PyTorch, OpenCV	Core development
Software	YOLO (Ultralytics), SAM 2, Depth Anything V2, COLMAP	Detection, labelling help, depth, 3D reconstruction
Software	CVAT or Label Studio	Labelling
Software	FastAPI, Next.js, Leaflet, PostgreSQL with PostGIS	Dashboard and map
Data	Team's own route videos	Main training and test data
Data	WireSegHR	Pre-training the cable model
Data	OpenStreetMap	Road lines for GPS correction
Data	Mapillary street images of Sri Lanka, where available	Extra frames; check the licence first

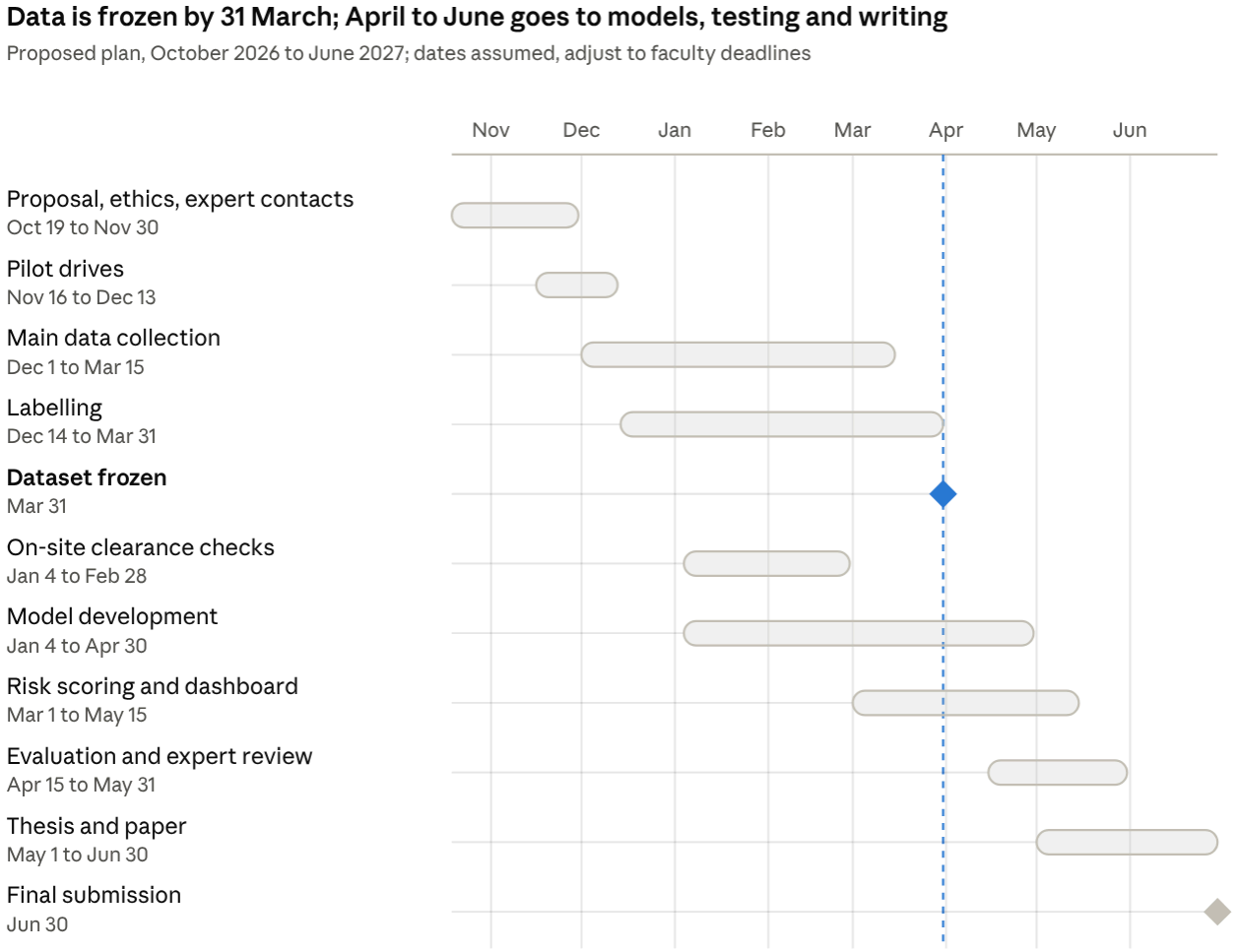
Team Roles

Each member owns one research question end to end, so everyone has a clear thesis chapter. All three share route drives, labelling and writing.

Member	Owns	Key outputs	Research question
Member A: trees	Tree detection and tracking, species, condition signs, fall-risk model	Tree detector, condition models, fall-risk score	3 (and the tree part of 1)

Member	Owns	Key outputs	Research question
Member B: cables and clearance	Cable segmentation and type, 3D reconstruction, clearance estimation, laser ground truth	Cable model, clearance method, error report	2 (and the cable part of 1)
Member C: data and dashboard	Capture protocol, GPS and sensor sync, frame selection, privacy blurring, map matching, foreign-vs-local experiments, dashboard	Clean dataset, transfer results, web map	4, plus whole-system testing

Timeline



project timeline · 9 phases, 2 milestones

Filming and labelling run through the drier months, and model work overlaps labelling so results are ready for evaluation in April and May.

Risks, Ethics and Safety

The biggest risks are safety near traffic and live wires, so no one films while driving and no one goes near a line.

Risk	Mitigation
Filming from a moving vehicle	Only a passenger operates the phone; mounts are checked before every drive
Live power lines	Never touch, climb or approach a line; clearance is measured from the ground with the rangefinder; any close-range work only with LECO or CEB crews
Privacy of people and vehicles in video	Faces and number plates blurred automatically before storage; only blurred frames shared or published
Ethics approval	Plan submitted to the faculty ethics review before any data collection
Expert time for labels	A tree expert (Forest Department, botany lecturer or arborist) and a utility contact lined up in the first month
Few high-risk trees in the data	Include routes with known problem trees; oversample them in training
Monsoon delays	Film mostly from December to March, usually the drier months in the Western Province; keep a buffer
Wrong clearance estimates	The tool is presented as a screening aid; field crews confirm before any trimming or felling

References

Sri Lankan context

- Daily Mirror (2023). Remove trees at risk of falling on roads: Minister.
- Daily FT (2025). Over 50,000 power outages reported amid monsoon rains.

Related studies

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Datasets

- Chiu, M. T., Zhang, X., et al. (2023). Automatic high resolution wire segmentation and removal (WireSegHR). CVPR.
- Abdelfattah, R., Wang, X., Wang, S. (2020). TTPLA: an aerial-image dataset for detection and segmentation of transmission towers and power lines.

Research notes and extra sources: Research notes

Research notes

Checked Oct 14, 2026. The biggest find: Sri Lanka already sets legal clearance distances between trees and lines, so the system can flag breaches against official numbers. It must also tell insulated low-voltage cables from bare ones, because their limits differ by up to 18 times.

Official clearance distances

The PUCSL wayleave guideline (version 4.0, January 2020) sets these distances from trees to overhead lines (PUCSL, 2020).

Line	Vertical clearance	Horizontal clearance
Low voltage, bare	2.7 m	1.5 m
Low voltage, insulated	0.15 m	0.15 m
11 kV	2.7 m	1.5 m
33 kV	3.7 m	2.9 m

When a tree is too close, the utility serves a tree notice. The occupier must comply or object within 7 working days; otherwise the utility may cut it under supervision (PUCSL, 2020).

PUCSL issued this guideline under the Sri Lanka Electricity Act, No. 20 of 2009, which the Sri Lanka Electricity Act, No. 36 of 2024 repeals. Confirm with PUCSL that these values still apply.

Evidence of the problem

Manual lists of risky trees still miss trees that fall, which is the gap a repeatable screening tool fills.

Date	Event	Source
4 Jun 2026	A eucalyptus tree fell on a moving three-wheeler at Mapakanda, Nawalapitiya, killing three people	Daily Mirror

Date	Event	Source
30 May 2025	About 50,000 electricity breakdowns in the 24 hours to 8 pm, announced by CEB, during monsoon rain and wind; fallen trees blocked major Colombo roads (reported 31 May)	Daily FT
21 Jun 2024	Colombo Municipal Council says fresh research with NBRO is under way to identify non-risky trees; recent bad weather mostly brought down trees not previously flagged as risky	Daily Mirror
17 May 2024	The council marks about 200 old, large trees in Colombo for removal	Daily Mirror
18 Dec 2023	The council reports 558 dangerous century-old trees	Daily Mirror
6 Oct 2023	The State Minister of Defence orders removal of roadside trees at risk of falling	Daily Mirror

The council and NBRO study is the closest local work: find out what it produced before the proposal, and treat NBRO as a possible partner.

Expert checklist to mirror

Label trees with the same categories experts already use, so the AI's output can be checked directly against an expert rating. The ISA Basic Tree Risk Assessment form records (ISA):

- **Site and targets:** slope, wind exposure, recent site changes, and what or who could be hit.
- **Tree condition:** crown health and density, live crown ratio.
- **Defects:** dead wood, cracks, decay, cavities, lean and included bark, on branches, trunk and roots.
- **Ratings:** likelihood of failure (Improbable to Imminent), likelihood of impact (Very low to High), and consequences (Negligible to Severe).
- **Result:** Low, Moderate, High or Extreme risk; each tree takes its highest rating.

Model and data options

Most model parts have a pretrained starting point; telling insulated from bare cables has none, so it becomes a new sub-task.

Need	Option	Use
Metric depth from one frame	UniDepth (2024) predicts metric 3D points without camera calibration and was tested zero-shot on driving datasets	Cross-check for tree crowns; too coarse for thin wires
Cable segmentation	WireSegHR (2023), TTPLA (aerial), and WireSeg-32K (2026; 32,000 synthetic images with masks and depth maps; check it is released)	Pre-train the cable model
Tree detection and species	Auto Arborist: about 4.5 million tree records from 23 cities, with aerial and street-level images	Pre-train the tree models
Insulated vs bare cable	No dataset found	New labels and a new classifier, needed because the clearance limits differ
Faster labelling	Open-vocabulary detectors such as Grounding DINO, plus SAM 2, draft outlines from the word "tree"	Team corrects drafts instead of drawing from scratch

Partners and checks

Line these up in the first month; expert and utility time is the slowest input.

Who	Ask for
Colombo Municipal Council and NBRO	Results of their 2024 tree study; lists of trees flagged or removed
LECO and CEB	How to tell insulated from bare lines; sample routes to confirm cable types; outage locations
PUCSL	Confirmation that the wayleave clearances are current
Botany lecturer or Forest Department	Species labels and expert risk ratings

Who	Ask for
UoM Civil Engineering surveying lab	Loan of a total station for ground-truth distances
University of Sri Jayewardenepura, Department of Geography	Their 2026 Colombo tree-risk map, to compare with this project's fall-risk scores

- [ ] Ask the council and NBRO whether their tree study can be shared
- [ ] Confirm with LECO which test routes use insulated cables
- [ ] Check whether WireSeg-32K is publicly released
- [ ] Request Auto Arborist access terms
- [ ] Book the total station for the January field checks

Sources

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